ABES ENGINEERING COLLEGE, GHAZIABAD

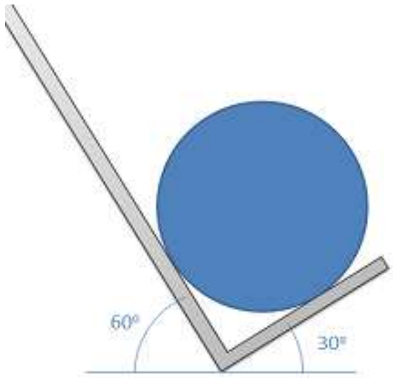
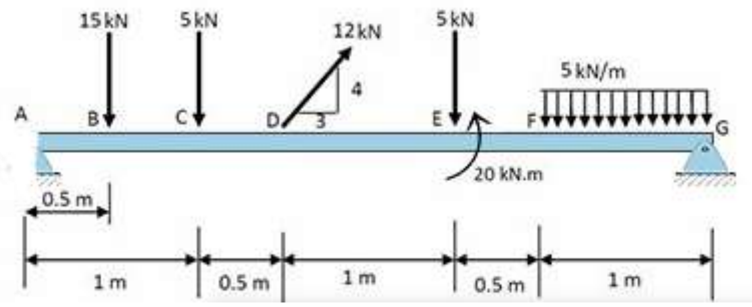
Department of Applied Sciences and Humanities

Basics of Mechanical and Automation Engineering (25ME101)

Assignment-1 Unit-I (Introduction to Mechanics and Strength of Materials)

Date of sharing: September 17, 2025 **Last Date of Submission: September 22, 2025**

Section : CSE_21 & 29

S. No.	CO	Questions	K Level
1	CO1	A roller of weight 233 kN rests in an L-shaped support-frame as shown in the figure. Determine the contact reactions at all the points of contact. Weight of the roller is 739 N. 	K3
2	CO1	A beam AG is subjected to various forces as shown in the figure. End A is a *simple support* and end G is *hinged*. Find the support reactions at these supported ends. 	K3
3	CO1	A square bar 55 mm x 55 mm is subjected to a compressive load of 600 KN. The contraction over 200 mm length is 0.45 mm and increase in thickness is 0.03 mm. Calculate the value of the four elastic constants of the material.	K3
4	CO1	Draw the stress strain diagram of mild steel and explain various salient points and regions on it.	K2
5	CO1	Derive the relation between Young's Modulus and Bulk Modulus	K2